

CSCE 585: Machine Learning Systems: Project Proposal

Eviction-Based Memory Management for 3D Gaussian Splatting SLAM

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September 8, 2026

1 Introduction

Online 3D Gaussian Splatting (3DGS) SLAM produces detailed maps and accurate tracking, but its map grows as Gaussians are added. On edge hardware, this can exhaust VRAM and make latency unpredictable. We will study whether a locality-aware active map can meet a strict memory budget while retaining tracking and reconstruction quality.

2 Problem Statement

Our question is: Which online eviction policy best balances peak VRAM, throughput, and SLAM/reconstruction quality under a fixed memory budget? We will use public TUM-RGBD sequences for trajectory accuracy and Replica for indoor reconstruction quality, including their supplied calibration and ground truth. The target platform is an NVIDIA Jetson AGX Orin.

3 Technical Approach

We will modify MonoGS [1] with a configurable VRAM budget and locality-aware eviction queue. When an allocation would exceed the budget, Gaussians will be ranked by recent visibility, rendered-pixel contribution, age, opacity, and spatial redundancy. We will compare (1) least-recently-visible eviction, (2) low-contribution visibility-aware culling, and (3) voxel-grid redundancy pruning. The ML component is online 3DGS optimization; our contribution is explicitly controllable memory use.

4 Evaluation

At 4 GB and 8 GB budgets, we will compare each policy with the unmodified baseline. Metrics are peak VRAM, out-of-memory events, FPS, latency, ATE RMSE, PSNR, and SSIM. We will report VRAM-over-time traces, budget-versus-ATE/FPS curves, and a sequence summary table. We expect visibility-aware culling to lose less quality than age-only eviction, while voxel pruning may lose fine geometry.

5 Related Work

MonoGS [1] and GS-SLAM [2] demonstrate online 3DGS SLAM. DiskChunGS [3] manages active memory with out-of-core chunks, while MemGS [4] uses voxel-space merging. PUP 3D-GS [5] and SafeguardGS [6] develop fidelity-preserving pruning scores. Our focus is lightweight online eviction under fixed VRAM budgets on an edge SLAM pipeline.

6 References

- [1] H. Matsuki et al., “Gaussian Splatting SLAM,” Proceedings of CVPR, 2024.
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- [3] C. Feldmann et al., “DiskChunGS: Large-Scale 3D Gaussian SLAM Through Chunk-Based Memory Management,” IEEE Robotics and Automation Letters, 2026.
- [4] Y. Bai et al., “MemGS: Memory-Efficient Gaussian Splatting for Real-Time SLAM,” arXiv:2509.13536, 2025.
- [5] A. Hanson et al., “PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting,” Proceedings of CVPR, 2025.
- [6] Y. Lee et al., “SafeguardGS: 3D Gaussian Primitive Pruning While Avoiding Catastrophic Scene Destruction,” arXiv:2405.17793, 2024.